

Thomas Falconer

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EDUCATION

Technical University of Denmark

Ph.D. Machine Learning & Economics

Kgs. Lyngby, Denmark

Jun. 2022 – Jun. 2025

University College London

M.Sc. Data Science (Distinction, Top in Year)

London, United Kingdom

Sep. 2019 – Nov. 2020

University of Amsterdam

B.Sc. Operations Research (Erasmus Exchange)

Amsterdam, Netherlands

Sep. 2016 – Jun. 2017

Heriot-Watt University

B.Eng. Engineering (First Class, Top in Year)

Edinburgh, United Kingdom

Sep. 2014 – Jun. 2019

EXPERIENCE

Quantitative Researcher (Algorithmic Trading)

Jun. 2025 – Present

InCommodities

Aarhus, Denmark

- Develop probabilistic machine learning (ML) models to forecast short-term European power prices.
- Apply (stochastic) programming methods to maximize portfolio PnL under uncertainty.
- Conduct rigorous backtesting and statistical validation to assess model reliability and strategy performance.
- Build real-time dashboards to monitor algorithms, data quality, and model drift.
- Implement CI/CD pipelines for automated model retraining, validation, and production deployment.

Machine Learning Engineer (Contract)

Jun. 2022 – Jun. 2025

Freelance (during Ph.d.)

Copenhagen, Denmark

- Designed and deployed end-to-end ML solutions for clients in energy and finance, including data ingestion, feature engineering, and model development.
- Developed maintainable codebases with testing, packaging, and documentation to ensure reliable ML pipelines.

Data Scientist

Dec. 2020 – Jun. 2022

Arenko Group

London, United Kingdom

- Designed and productionised ML models for electricity market price forecasting.
- Implemented ensembles of forecasting algorithms, including Gaussian processes, neural networks and gradient boosting, plus simpler statistical models.
- Developed derivative-free optimization models combining population-based metaheuristics (particle swarm, genetic algorithms) and Monte Carlo simulation to trade battery assets over discontinuous, stochastic surfaces.
- Built MLOps pipelines in AWS, including feature store, model versioning (MLflow), low-latency model serving (FastAPI), data engineering and orchestration (Prefect) and digestion (RDBMS) in a microservice framework.
- Created interactive visualisations of market opportunities (matplotlib, Plotly, Dash, Streamlit).
- Mentored junior data scientists and helped grow the data science team.

Machine Learning Researcher (Intern)

May 2019 – Aug. 2019

Invenia Labs

Cambridge, United Kingdom

- Developed ML pipelines to forecast nodal price spreads for virtual bidding across US ISO markets.
- Applied graph neural networks to model grid topology for congestion forecasting and optimal power flow.
- Co-authored NeurIPS workshop paper and IEEE journal publication on ML for power systems.

TECHNICAL SKILLS

Languages: Python, Julia, SQL, Go, C, C++, Bash

Quality: unit testing, packaging, documentation, coding standards

Analysis: pandas, Polars, NumPy; experiment design, statistical testing, feature engineering, model explainability

Modelling: scikit-learn, LightGBM, PyTorch, TensorFlow; Bayesian inference, uncertainty quantification, validation

Optimization: Pyomo, CVXPY, Gurobi; stochastic programming, constrained optimization, DRO, metaheuristics

MLOps: Docker, FastAPI, MLflow, Prefect, Argo; ETL, versioning, model serving, orchestration, monitoring

Infrastructure: AWS (S3, EC2, Lambda), PostgreSQL, CI/CD, logging (Humio), package distribution (Gemfury)

Visualization: matplotlib, Plotly, Dash, Streamlit